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Review Article

The role of serratus anterior plane and pectoral nerves blocks in cardiac surgery, thoracic surgery and trauma: a qualitative systematic review

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Summary

Serratus anterior plane and pectoral nerves blocks are recently described alternatives to established regional anaesthesia techniques in cardiac surgery, thoracic surgery and trauma. We performed a systematic review to establish the current state of evidence for the analgesic role of these fascial plane blocks in these clinical settings. We identified relevant studies by searching multiple databases and trial registries from inception to June 2019. Study heterogeneity prevented meta-analysis and studies were instead qualitatively summarised and stratified by type of surgery and comparator. We identified 51 studies: nine randomised control trials; 13 cohort studies; 19 case series; and 10 case reports. The majority of randomised controlled trials studied the serratus anterior plane block in thoracotomy or video-assisted thoracoscopic surgery, with only two investigating pectoral nerves blocks. The evidence in thoracic trauma comprised only case series and reports. Results indicate that single-injection serratus anterior plane and the pectoral nerves blocks reduce pain scores and opioid consumption compared with systemic analgesia alone in cardiothoracic surgery, cardiac-related interventional procedures and chest trauma for approximately 6–12 h. The duration of action appears longer than intercostal nerve blocks but may be shorter than thoracic paravertebral blockade. Block duration may be prolonged by a continuous catheter technique with potentially similar results to thoracic epidural analgesia. There were no reported complications and the risk of haemodynamic instability appears to be low. The current evidence, though limited, supports the efficacy and safety of serratus anterior plane and the pectoral nerves blocks as analgesic options in cardiothoracic surgery.

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Introduction

Cardiac surgery, thoracic surgery and trauma are associated with significant pain, respiratory compromise and a consequent risk of pulmonary morbidity. Effective multimodal analgesia is the cornerstone of management, with regional anaesthesia widely viewed as an essential component of an optimal analgesic regimen. Fascial plane blocks of the anterolateral chest wall have become popular

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as alternatives to more-established techniques such as thoracic epidural analgesia and thoracic paravertebral blockade. The fascial plane block techniques that are most commonly practiced include the serratus anterior plane block and pectoral nerves (PECS) blocks. There are two described variants of the serratus anterior plane block: the superficial serratus anterior plane block and the deep serratus anterior plane block, which involve local anaesthetic injection superficial or deep to the serratus anterior muscle, respectively. Both target the anterolateral branches of the thoracic intercostal nerves [1, 2]. The term PECS block similarly encompasses two separate techniques. The PECS 1 block involves local anaesthetic injection between pectoralis major and minor muscles and blocks the lateral and medial pectoral nerves as well as the intercostobrachial nerve [2, 3]. The PECS 2 block is a combination of a PECS 1 block and superficial serratus anterior plane block that provides more extensive coverage of the chest wall and the axilla [2, 4].

Serratus anterior plane and PECS blocks are most commonly associated with breast surgery and their efficacy here was recently established [4]. However, there have been increasing reports of the use of these blocks for chest wall analgesia in cardiac surgery, thoracic surgery and trauma. We therefore undertook a systematic review of the published literature on serratus anterior plane and PECS blocks in the setting of analgesia for cardiothoracic surgical procedures and trauma in order to better define their clinical role and to determine the quality of evidence that currently exists for these applications.

Methods

We included any study reporting the application of PECS 1, PECS 2 or serratus anterior plane block in thoracic trauma, or in elective cardiac or thoracic surgical procedures. Eligible study types included: randomised controlled trials (RCTs); prospective non-randomised comparative studies; retrospective comparative and cohort studies; prospective observational cohort studies; and case series or reports.

We conducted a systematic review using the methodology described by the Cochrane Handbook [5] and the Cochrane methodological expectations of Cochrane intervention reviews [6]. Search results were reported according to the preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines [7, 8]. The peer review of electronic search strategies (PRESS) guideline was followed in peer-reviewing the search strategies [9], drawing on the PRESS 2015 guideline evidence-based checklist to avoid potential search errors. The review protocol was prospectively registered on PROSPERO.

We searched the following databases: MEDLINE; Medline In-Process/ePubs; Embase; Cochrane Central Register of Controlled Trials; and Cochrane Database of Systematic Reviews. We also searched the PubMed database (National Library of Medicine) for non-Medline citations, Scopus, Web of Science and the clinicaltrials.gov (National Institutes of Health) and World Health Organization's International Clinical Trials Registry Platform trials registries. All databases and trial registries were searched from inception until 26 June 2019. Preliminary searches were conducted and full-text literature was mined for potential keywords and appropriate controlled vocabulary terms. The search strategy concept blocks were built on the topics of 'PECS blocks' or 'serratus blocks' and 'surgery of cardio or cardiac or thoracic or ribs or torso' and 'studies' limited to human. No publication type, status or language restrictions were applied. The search strategies for all databases are described in the Supporting Information (Appendix S1). Potentially eligible studies were identified by screening the title and abstract of all publications identified by the literature search. The full-text of these studies was then obtained for further evaluation. Screening and evaluation for inclusion in the review were independently and separately performed by two authors (JJ, EM). Any disagreements on inclusion were resolved by discussion with a third author (KC).

Data on the following outcomes were extracted from RCTs and non-randomised comparative trials: resting and dynamic pain scores; intra-operative and postoperative analgesic requirements; time to first postoperative analgesic administration; duration of postoperative ventilatory support; changes in haemodynamic parameters; length of intensive care unit (ICU) stay; length of hospital stay; patient satisfaction; adverse effects; and complications. All other studies, including RCTs and cohort studies published in abstract form only, were qualitatively summarised to provide information on: potential indications; analgesic effect; technical aspects; advantages; and disadvantages of these regional anaesthesia techniques.

We assessed the methodological quality of included RCTs using the Cochrane Collaboration's risk of bias summary tool [10] and that of non-randomised cohort studies using the Newcastle-Ottawa Scale [11], which is based on an evaluation of study group selection process, comparability of groups and ascertainment of the outcomes of interest. We planned a priori to pool data from studies that used the same intervention and comparator, and that assessed similar outcomes at comparative time-points and intervals. We also planned to subsequently perform a

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quantitative random-effects meta-analysis, with standardised mean differences for continuous outcomes and risk ratios for binary outcomes using aggregate data. Data analysis was planned to be stratified by type of intervention and comparator. Where the study design or available data did not allow quantitative analysis, we performed a qualitative analysis and summary of study findings instead. The term 'significant' refers to statistically significant results as reported by each study, and makes no assumptions as to clinical significance, unless otherwise stated. All pain scores are reported on a 0–10 scale, unless otherwise stated.

The level of available evidence for PECS 1, PECS 2 and serratus anterior plane block in specific clinical applications was graded in accordance with the 2011 Oxford Centre for Evidence-based Medicine levels of evidence document as follows: level 1, evidence from either n-of-1 randomised trials or systematic reviews of randomised trials; level 2, evidence from randomised trials or observational studies with dramatic effect; level 3, evidence from non-randomised controlled cohort or follow-up studies; level 4, evidence from case series, case-control studies or historically controlled studies; and level 5, mechanism-based reasoning only [12].

Results

The results of the search strategy are presented in Fig. 1. Fifty-one studies were identified for inclusion in this review, of which nine were RCTs [13–21] involving a total of 637 patients (see also Supporting Information, Table S1). Eight RCTs evaluated the serratus anterior plane block in the setting of video-assisted thoracoscopic surgery (five studies [13–17]) and thoracotomy (three studies [18–20]). Five of these compared single-injection serratus anterior plane block with systemic analgesia alone [13–17]; one compared it with thoracic paravertebral blockade and with systemic analgesia alone [18]; and one compared it with PECS 2 blocks and with intercostal nerve blocks [19]. Only one study evaluated a continuous catheter technique, comparing it with thoracic epidural analgesia [20]. Two RCTs evaluated PECS blocks. The first compared PECS 2 blocks, serratus anterior plane block and intercostal nerve blocks in paediatric patients undergoing thoracotomy for cardiac surgery [19]. In the second, PECS 2 blocks were compared with systemic analgesia in adult patients undergoing median sternotomy for cardiac surgery [21]. There were no RCTs involving thoracic trauma.

There were 42 non-randomised studies: 13 comparative cohort studies [22–34]; 19 case series [35–53]; and 10 case reports [54–63]. Six of the cohort studies [25–29, 33]; 14 of the case series [35, 36, 38–43, 46–49, 52, 53]; and seven of the case reports [55, 57–62] were available only as abstracts or posters from conference proceedings. Twenty-nine non-randomised studies described the use of serratus anterior plane block in thoracic trauma and in surgical procedures including lung resection and minimally invasive cardiac surgery: nine comparative cohort studies [22, 23, 25, 26, 29–32, 34]; 13 case series [36–42, 45, 47, 48, 50, 51, 53]; and eight case reports [54, 55, 57–60, 62, 63]. Fourteen non-randomised studies evaluated PECS blocks in a wide range of cardiothoracic applications, including: thoracotomy; sternotomy; transcatheter cardiac procedures; cardiac device placement; and thoracic trauma. These comprised: four comparative cohort studies [24, 27, 28, 33]; seven case series [35, 39, 43, 44, 46, 49, 52]; and two case reports [56, 61].

Risk of bias in the nine RCTs was assessed to be generally low, with the exception of two studies [16, 18] (Fig. 2). The seven cohort studies published in the peer-reviewed literature [22–24, 30–32, 34] had a median (IQR [range]) score of 7 (7–7 [6–8]) on the 9-point Newcastle-Ottawa Scale, whereas the six cohort studies published as abstracts or posters [25–29, 33] had a median (IQR [range]) score of 6 (5.25–6 [2–6]) (see also Supporting Information, Table S2).

Due to the small number of comparative studies and the heterogeneity in surgical population, intervention technique and comparators, meta-analysis was not undertaken and a qualitative synthesis grouped by type of surgery and comparator was performed instead. The overall findings are summarised in Tables 1 and 2.

Thoracotomy

We identified three RCTs involving 238 patients [18–20] and five retrospective cohort studies involving 375 patients [22, 23, 25, 30, 34] that investigated the serratus anterior plane block for thoracotomy surgery. Serratus anterior plane block was compared with: multi-level intercostal nerve blocks in two studies [19, 34]; thoracic paravertebral blockade in two studies [18, 30]; systemic analgesia alone in three studies [18, 23, 30]; thoracic epidural analgesia in one study [20]; and continuous wound infiltration in one study [22]. Three of these studies had more than two treatment arms: Saad et al. compared serratus anterior plane block vs. thoracic paravertebral blockade vs. systemic analgesia [18]; Moll et al. compared single-injection serratus anterior plane block vs. continuous serratus anterior plane block vs. single-injection thoracic paravertebral blockade vs. systemic analgesia alone [30]; and Kaushal et al. compared serratus anterior plane block vs. PECS 2 block vs. intercostal nerve blocks [19]. The work done by Kaushal et al. was the only comparative study of the PECS 2 block in thoracotomy. Findings are summarised in the Supporting Information (Table S3).

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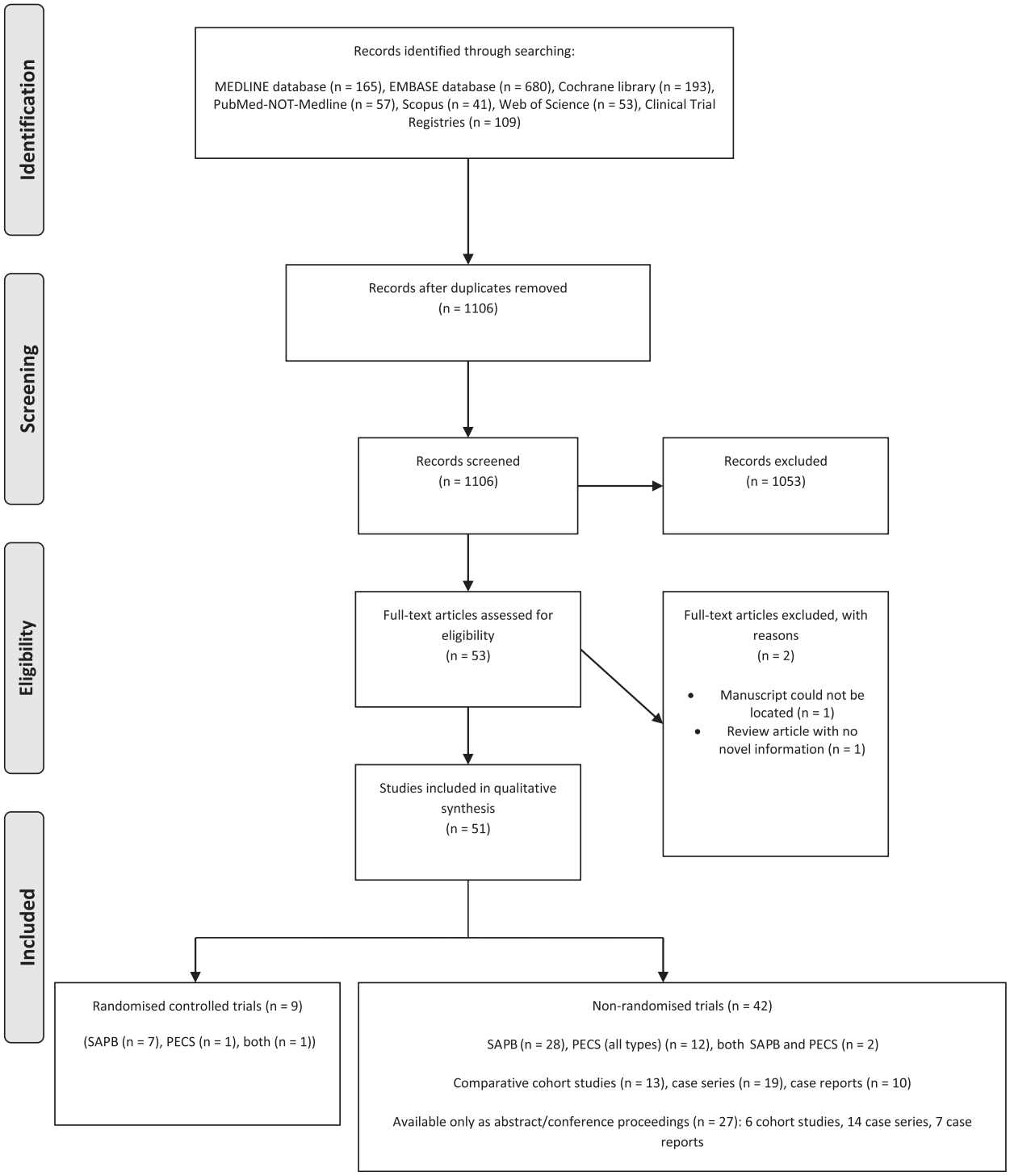


Figure 1 PRISMA flow diagram of studies in this systematic review. SAPB, serratus anterior plane block; PECS, pectoral nerves.

Serratus anterior plane block and PECS block vs. systemic analgesia

There is level 2 evidence from a three-armed RCT of 90 patients by Saad et al. that a pre-incisional single-injection serratus anterior plane block improves early post-thoracotomy analgesia compared with systemic analgesia alone [18]. The serratus anterior plane block group had significantly lower pain scores up to 9 h postoperatively

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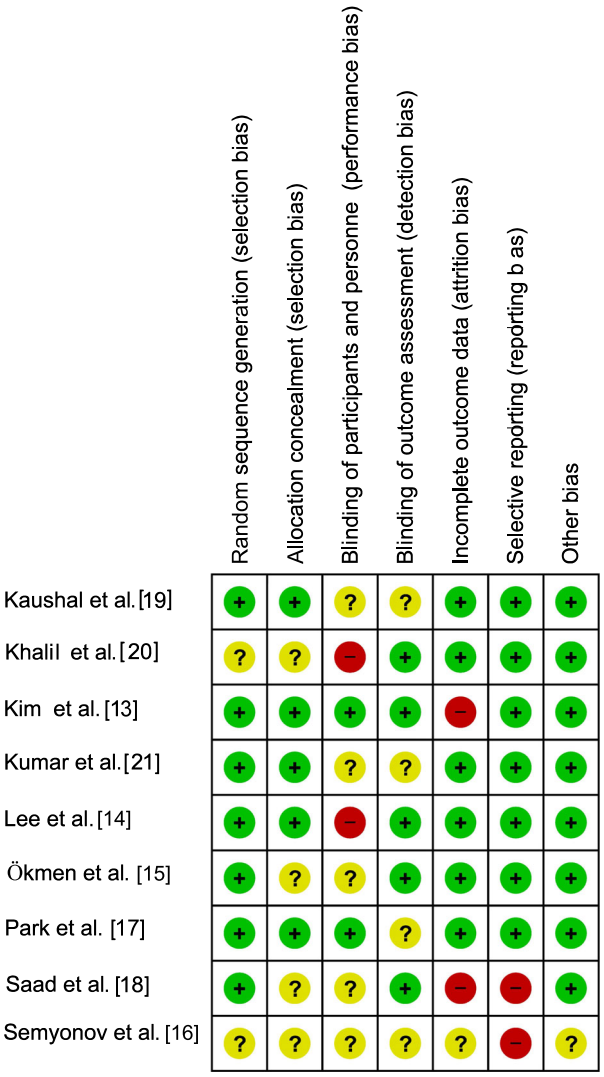


Figure 2 Cochrane risk of bias tool for included randomised control trials.

(median [range] scores ranging from 1 [0–3] to 3 [0–5] vs. 2 [0–5] to 4 [2–9]; IQR not reported); and lower 24-h opioid requirements. This is supported by level 3 evidence from a retrospective study comparing postoperative deep serratus anterior plane block with systemic opioids alone in 40 patients [23]. At all time-points up to 24 h postoperatively, the serratus anterior plane block group had significantly lower pain scores (mean (SD) scores ranging from 1.8 (1.1) to 5.2 (0.8) vs. 3.3 (1.0) to 6.3 (1.5)) and opioid consumption (reductions ranging from 17% to 24%) than systemic opioids alone, respectively. Six level 4 evidence studies (a retrospective cohort study abstract [25]; three case series [36–38]; and two case reports [54, 55]) have also described the successful use of serratus anterior plane block for post-thoracotomy analgesia.

In contrast, a retrospective cohort study of 197 patients found that cumulative 72-h opioid consumption was similar between patients who received single-injection or continuous serratus anterior plane block compared with systemic analgesia alone (level 3 evidence) [30]. However, there were no data for shorter time intervals nor was there mention of whether multimodal systemic analgesia was used.

There is currently only one described case of PECS 2 block for post-thoracotomy analgesia as part of a case series that principally described its use for rib fracture analgesia (level 4 evidence) [49].

Serratus anterior plane block and PECS block vs. intercostal nerve blocks

There is level 2 evidence from a three-armed RCT of 108 paediatric patients by Kaushal et al. that a single-injection deep serratus anterior plane block provides superior post-thoracotomy analgesia compared with five-level intercostal nerve blocks [19]. All patients received pre-incisional blocks and postoperative intravenous acetaminophen 8-hourly. Significantly lower pain scores were observed in the serratus anterior plane block group between 6 h and 12 h post-extubation (mean (SD) scores ranging from 2.2 (0.5) to 3.0 (0.2) vs. 2.7 (0.5) to 3.9 (0.4), respectively). None of the patients required rescue opioid in the first 4 h post-extubation, but opioid consumption over 12 h was 44% higher in the intercostal nerve block group. This is consistent with level 3 evidence from a retrospective cohort study of 42 adult patients who reported that, compared with surgically performed intercostal nerve blocks, a postoperative serratus anterior plane block resulted in significantly lower pain scores at all time-points up to 24 h (mean scores ranging from 1–4 vs. 3–5, respectively; SD not reported) and a 17% reduction in 24-h opioid consumption [34].

The RCT by Kaushal et al. also provides level 2 evidence for superior post-thoracotomy analgesia following PECS 2 block compared with five-level intercostal nerve blocks [19], with significantly lower mean (SD) pain scores between 6 h and 10 h after extubation (scores ranging from 2.2 (0.5) to 3.0 (0.5) vs. 2.7 (0.5) to 3.9 (0.4), respectively), but not at 12 h. Cumulative opioid consumption in the PECS 2 group was 23% lower in the first 12 h post-extubation.

Serratus anterior plane block vs. PECS 2 block

The RCT by Kaushal et al. is currently the only study comparing serratus anterior plane and PECS 2 blocks and suggests that they provide similar post-thoracotomy analgesia (level 2 evidence) [19]. There was no difference in

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Table 1 Summary of studies and the highest level of evidence according to the Oxford Centre for Evidence-Based Medicine levels of evidence [12] for serratus anterior plane blocks (SABP).

Indication	Comparator	Highest level of evidence	Available studies	Outcomes
Thoracotomy	Systemic analgesia	Level 2	one RCT (n = 90) [18] 2 retrospective cohort studies (n = 40) [23]; (n = 197) [30] three case series (n = 25) [36]; (n = 7) [37]; (n = 4) [38] two case reports [54,55]	Lower pain scores and opioid requirements with SAPB [18]
	Intercostal nerve blocks (five-level single-injection)	Level 2	one RCT (n = 108) [19] one retrospective cohort study (n = 42) [34]	Lower pain scores and opioid requirements with SAPB [19]
	Thoracic paravertebral block (single-injection)	Level 2	one RCT (n = 90) [18] one retrospective cohort study (n = 197) [30]	Similar analgesic effect up to 9–12 h. Superior pain scores and opioid consumption at 12–24 h with TPVB. Fewer episodes of hypotension associated with SAPB [18]
	Thoracic epidural	Level 2	one RCT (n = 40) [20] one retrospective cohort study (n = 50) [25]	Similar 2-hourly pain scores (except at 14 h, 16 h and 22 h) and similar opioid requirements. Higher MAPs and fewer hypotensive episodes associated with SAPB [20]
	PECS 2 block	Level 2	one RCT (n = 108) [19]	Comparable pain scores and opioid requirements between SAPB and PECS blocks [19]
	Continuous wound infusion	Level 4	one retrospective cohort study (n = 46) [22]	Lower early pain scores (but no difference at time-points 6 h, 24 h and 48 h) and 50% reduction in 48-h cumulative opioid consumption with SAPB [22]
Video-assisted thoracoscopic surgery	Systemic analgesia	Level 2	five RCTs (n = 85) [13] (n = 46) [14], (n = 40) [15], (n = 104) [16], (n = 84) [17]. one retrospective cohort study (n = 34) [31]	Lower pain scores and 21–46% reductions in opioid consumption [13–17] with SAPB. No significant differences in heart rate and MAP [13, 16]
Thoracic trauma	Not applicable	Level 4	SAPB only one retrospective cohort study (n = 50) [29] five case series (n = 5) [40], (n = 9) [47], (n = 11) [48], (n = 5) [51], (n = 17) [53] five case reports [57,59,60,62,63] PECS and SAPB one case series (n = 2) [39] one case report [58]	SAPB reduced pain scores in patients with rib fractures
Cardiac interventional procedures	Not applicable	Level 3	one cohort study (n = 20) [26] two case series (n = 12) [45]; (n = 19) [50]	SAPB used as sole anaesthetic technique instead of GA [26]. SAPB used in combination with GA [50]; and as a standalone anaesthetic technique [26, 45] to facilitate cardiac device implantation
Minimally invasive repair of pectus excavatum (Nuss procedure)	Systemic analgesia	Level 4	one retrospective cohort study (n = 95) [32]	Reduced 24-h opioid requirement and no difference in median postoperative pain scores with SAPB

(continued)

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Table 1 (*continued*)

Indication	Comparator	Highest level of evidence	Available studies	Outcomes
Portacath insertion	Not applicable	Level 4	one case series (n = 8) [42]	SAPB provided effective postoperative analgesia and eliminated opioid requirements in paediatric patients
Latissimus dorsi tendon transfer surgery	Thoracic paravertebral block	Level 4	one case series (n = 4) [41]	Lower pain scores and reduction in opioid requirements with SAPB

GA, general anaesthesia; MAP, mean arterial blood pressure; PECS, pectoral nerve blocks; RCT, randomised control trial; TPVB, thoracic paravertebral block.

12-h opioid consumption following extubation. Mean (SD) pain scores were also similar up to 10 h post-extubation, with a significant difference observed only at 12 h in favour of the serratus anterior plane block group vs. PECS 2 block group (3.0 (0.2) vs. 3.8 (0.4), respectively).

Serratus anterior plane block vs. thoracic paravertebral blockade

There is level 2 evidence from the three-armed RCT by Saad et al. that early post-thoracotomy analgesia is similar between a single-injection superficial serratus anterior plane and T5 thoracic paravertebral block, but thoracic paravertebral blockade is associated with lower pain scores and opioid consumption at 12–24 h [18]. There were no significant differences in intra-operative opioid consumption or median (range) scores pain scores up to 9 h postoperatively, but pain scores were significantly lower in the thoracic paravertebral block group than the serratus anterior plane block at 12 h and 24 h postoperatively (pain scores 0 (0–3) vs. 3 (0–5) at both time-points, respectively; IQR not reported). Twenty-four-hour cumulative opioid requirements were also lower, and 77% of patients required no rescue doses compared with 3% of patients in the serratus anterior plane block group. It is unclear if this is due primarily to a difference in duration of effect between the two techniques, or if other factors such as extent or intensity of block may also be at play. A retrospective cohort study found that both single-injection and continuous serratus anterior plane block were associated with higher 72-h opioid consumption (51% and 37% increase, respectively) when compared with a single-injection T4 thoracic paravertebral block (level 3 evidence) [30]. This was ascribed to the failure of the serratus anterior plane block to cover the thoracostomy incision in the lower thoracic wall. There are currently no comparative studies of continuous serratus anterior plane block and continuous thoracic paravertebral block.

Serratus anterior plane block vs. thoracic epidural analgesia

There is level 2 evidence from one RCT of 40 patients that continuous superficial serratus anterior plane block may provide similar post-thoracotomy analgesia compared with thoracic epidural analgesia in thoracotomy [20]. Postoperative median (IQR) pain scores measured every 2 h were similar between both groups except at 14 h, 16 h and 22 h when pain scores were higher in the serratus anterior plane block group than the thoracic epidural group (ranging from 2 (2–2) to 4 (3–4) vs. 1 (1–2) to 2 (1–2), respectively; range not reported). There was no significant difference in 24-h opioid consumption.

Serratus anterior plane block vs. continuous wound infiltration

There is level 3 evidence from a retrospective cohort study of 46 patients that single-injection deep serratus anterior plane block may be more effective than continuous wound infiltration for post-thoracotomy analgesia [22]. Berthoud et al. compared serratus anterior plane block performed at the end of surgery with continuous local anaesthetic infiltration for 48 h via a surgically inserted wound catheter. Median (IQR) pain scores were significantly lower in the serratus anterior plane block group than the infiltration group in the post-anaesthetic recovery unit (0 (0–4) vs. 5 (4–6), respectively; range not reported). Although pain scores were similar at 6 h, 24 h and 48 h, cumulative opioid consumption was significantly lower in the serratus anterior plane block group at all time-points, with up to a 50% decrease at 48 h postoperatively compared with the wound infiltration group. The duration of postoperative ventilatory support was similar; however, the serratus anterior plane block group had significantly decreased ICU (median (IQR) 24 (20–42) vs. 41 (25–69) h; range not reported) and hospital lengths of stay (median (IQR) 6 (5–8) vs. 8 (7–9) days; range not reported).

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Table 2 Summary of studies and the highest level of evidence according to the Oxford Centre for Evidence-Based Medicine Levels of Evidence [12] for pectoral nerves blocks (PECS).

Indication	Comparator	Highest level of evidence	Available studies	Outcomes with intervention
Thoracotomy	Intercostal nerve blocks (five-level single-injection)	Level 2	one RCT (n = 108) [19]	Lower pain scores and opioid requirements with PECS blocks
	SAPB	Level 2	one RCT (n = 108) [19]	Comparable pain scores and opioid requirements between PECS blocks and SAPB
Video-assisted thoracoscopic surgery	Not applicable	Level 4	one case series (n = 10) [35]	Analgesic benefit lasting 18–24 h with pain scores < 6, > 50% required no opioids [35]
Median sternotomy	Systemic analgesia	Level 2 evidence for PECS 2 block Level 4 evidence for PECS 1 block	one RCT (n = 40) [21] one retrospective cohort study (n = 112) [27]	Lower pain scores up to 18 h and fewer episodes of rescue analgesia with PECS blocks [21]. Reduced opioid consumption with PECS blocks [27]
Cardiac interventional procedures	Systemic analgesia	Level 3	two retrospective cohort studies (n = 32) [28], (n = 26) [33] three case series (n = 4) [43], (n = 3) [44], (n = 6) [46] one case report [56]	PECS 2 blocks used alone and in combination with GA to facilitate cardiac device implantation and transcatheter procedures. PECS 1 blocks provided effective analgesia in paediatric patients in combination with GA for cardiac device implantation
Thoracic trauma	Not applicable	Level 4	PECS only two case series (n = 14) [49]; (n = 2) [52] one case report [61] PECS and SAPB one case series (n = 2) [39] one case report [58]	Reduced pain scores in patients with rib fractures with PECS 2 blocks alone and in combination with SAPB
Portacath insertion	Local anaesthetic wound infiltration	Level 4	one retrospective cohort study (n = 25) [24]	No clinically significant difference in postoperative pain scores or opioid requirements of the five patients who received PECS 1 blocks
	Not applicable	Level 4	one case report [43]	Patient comfortable and required no opioids
Chest wall liposuction	Not applicable	Level 4	one case report [43]	Patient comfortable and required no opioids

GA, general anaesthesia; RCT, randomised control trial; SAPB, serratus anterior plane block.

Video-assisted thoracoscopic surgery

There is level 2 evidence from five RCTs involving 359 patients that single-injection serratus anterior plane block provides superior postoperative analgesia for up to 24 h after video-assisted thoracoscopic surgery compared with systemic analgesia alone [13–17]. Kim et al. [13] compared pre-operative superficial serratus anterior plane block with placebo block and observed significantly lower postoperative resting mean (SD) pain scores at 6 h (4.5 (1.9) vs. 6.1 (1.6), respectively) in the active intervention group, but not at 24 h or 48 h. Opioid consumption in the first 24 h was also significantly reduced by 21%. Patients who received serratus anterior plane block had higher quality of recovery-40 (QoR-40) scores on postoperative day one and two (mean difference of 8.5). There were no significant differences in intra-operative opioid requirements, pain

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scores when coughing or length of hospital stay. Park et al. [17] compared pre-operative deep serratus anterior plane block with no block and reported significantly lower median (IQR[range] pain scores in the intervention group at all time-points up to 24 h postoperatively (maximum scores ranging from 2 (2–3 [3–8]) to 5 (3–7 [2–10]) vs. 4 (3–7 [1–10]) to 6 (5–7 [2–10]), respectively), a 33% reduction in 24-h opioid requirements and greater satisfaction with analgesia. The length of hospital stay was similar in both groups. All patients in both these studies [13, 17] received postoperative multimodal analgesia.

Ökmen et al. [15] compared a postoperative deep serratus anterior plane block with no block in the absence of postoperative non-opioid analgesia. The serratus anterior plane block group had significant reductions in mean (SD) resting pain scores at time-points from 2 to 24 h postoperatively compared with no block (ranging from 2.0 (0.9) to 3.7 (0.9) vs. 3.4 (0.9) to 4.9 (0.6), respectively) and significantly lower cumulative opioid consumption at 6 h, 12 h and 24 h postoperatively (21–43% decrease).

Lee et al. [14] compared a pre-operative superficial serratus anterior plane block with no block and found that the intervention group had a 51% decrease in intra-operative remifentanil requirements, which was also associated with a more rapid emergence. Pain scores and postoperative opioid use were not reported.

Finally, Semyonov et al. [16] reported lower mean (SD) pain scores up to 8 h postoperatively in patients receiving a pre-incisional single-injection serratus anterior plane block compared with no block (ranging from 2.1 (1.3) to 3.6 (2.1) vs. 2.8 (1.6) to 6.6 (2.3), respectively). The intervention group also had significantly lower opioid requirements during the first hour in the recovery area (46% reduction) and less postoperative vomiting in the first 24 h (4% vs. 12%). It should be noted that this paper was graded at high risk of bias (Fig. 2).

There were no studies comparing serratus anterior plane block with other regional anaesthesia techniques. The PECS 2 block for postoperative analgesia after video-assisted thoracoscopic surgery was described in one retrospective observational study abstract (level 4 evidence) [35]. In this study of 10 patients, pain scores ranged between 0 and 6, and four patients required no intravenous opioids during the first 24 h postoperatively.

Median sternotomy

There are currently no studies of the serratus anterior plane block for analgesia following median sternotomy. There is level 2 evidence from one RCT of 40 adult patients that postoperative PECS 2 block is superior to systemic analgesia alone [21], significantly reducing mean (SD) pain scores at rest (ranging from 1.0 (1.0) to 1.9 (1.1) vs. 3.3 (1.0) to 4.5 (0.7), respectively) and on coughing (ranging from 1.7 (1.3) to 2.8 (1.4) vs. 4.1 (1.1) to 5.2 (0.9), respectively) at all time-points up to 18 h post-extubation. There was no difference in pain scores at 24 h. The PECS 2 group also had significantly lower opioid requirements than the control group, along with a significantly shorter time to extubation and higher peak inspiratory flow rates at all time-points up to 24 h.

There is level 4 evidence for the analgesic effect of PECS 1 block in midline sternotomy from a retrospective study abstract of 112 patients [27]. There were two intervention groups who received either bilateral PECS 1 blocks (n = 65) or bilateral PECS 1 with subcostal TAP blocks (n = 18). Both groups used significantly less opioid intra-operatively and postoperatively than patients receiving systemic analgesia only. There was no significant difference between the two intervention groups and there was no difference in length of hospital stay between the three groups.

Cardiac interventional procedures

There is currently only level 4 evidence to support the use of serratus anterior plane and PECS blocks in the context of cardiac interventional procedures (see also Supporting Information, Table S4). Two case series have described the successful implantation of internal cardiac defibrillators using a serratus anterior plane block without general anaesthesia [26, 45] and a third case series described its efficacy in combination with general anaesthesia [50]. One retrospective cohort study [28]; three case series [33, 44, 46]; and one case report [56] have described using PECS 2 blocks, either in combination with general anaesthesia or as the sole anaesthetic modality, for transcatheter cardiac procedures or cardiac device implantation.

Thoracic trauma

There is currently only level 4 evidence for the use of serratus anterior plane block or PECS blocks in thoracic trauma (see also Supporting Information, Table S5). Five case series [40, 47, 48, 51, 53] and five case reports [57, 59, 60, 62, 63] reported that single-injection or continuous serratus anterior plane block reduced pain scores in patients with rib fractures. Of note, this included a series of five patients with isolated posterior rib fractures who were successfully managed with a deep serratus anterior plane block catheter inserted in the mid-axillary line [63]. We identified one retrospective cohort study of 50 patients with

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rib fractures that reported equivalent analgesia between groups who received either thoracic epidural analgesia, thoracic paravertebral blockade or serratus anterior plane block; however, this was available only as an abstract [29].

Effective analgesia after rib fracture with PECS 2 blocks was described in one case report [61] and one case series of 12 patients [49]. Another case series reported the successful use of unspecified subpectoral interfascial plane blocks for analgesia facilitating tracheal extubation in patients with iatrogenic sternal and rib fractures secondary to cardiopulmonary resuscitation [52]. The combination of serratus anterior plane block and PECS 2 blocks has also been described in a case series of patients with both clavicular and rib fractures [39] and in one patient with rib fractures [58].

Other clinical indications

Five studies [24, 32, 41–43] described serratus anterior plane block and PECS blocks for thoracic analgesia for indications other than those already outlined above (level 4 evidence; see also Supporting Information, Table S6). In a retrospective cohort study of 95 patients of age 8–39 years undergoing minimally invasive pectus excavatum repair (Nuss procedure), bilateral pre-incisional single-injection superficial serratus anterior plane blocks resulted in significantly lower opioid requirements in the first 24 h compared with no block, with a median (IQR) patient-controlled demands of 23 (14–48) vs. 61 (31–136), respectively, and 15 (10–22) vs. 25 (15–42) doses delivered, respectively (ranges not reported) [32]. Median (IQR) pain scores were similarly low in both groups (0 (0–1) at all time-points; ranges not reported). In a case series of four patients undergoing latissimus dorsi tendon transfer with a brachial plexus block, pain scores and opioid requirements in the recovery area were lower in patients who received an additional serratus anterior plane block compared with those who received a thoracic paravertebral block instead [41]. The need for intra-operative opioids was eliminated by the use of pre-operative PECS 1 blocks in four paediatric patients undergoing liposuction for gynaecomastia, cardiac device implantation and portacath insertion under general anaesthesia [43]. A retrospective cohort study evaluating 25 paediatric patients undergoing portacath insertion or removal under general anaesthesia found similar pain scores and rescue analgesic requirements between those who received pre-operative PECS 1 blocks (five patients) or local anaesthetic wound infiltration [24]. Effective peri-operative analgesia was similarly provided by deep serratus anterior plane block in eight paediatric patients undergoing portacath insertion under general anaesthesia [42].

Adverse effects and complications

Serratus anterior plane or PECS blocks did not increase the risk of complications or adverse effects compared with control groups in any of the RCTs that evaluated these outcomes (level 2 evidence) [13, 14, 16, 17, 19]. There were no block-related complications identified in the reviewed literature, which included a total of 875 blocks performed in 785 patients. Specifically, there were no reports of: pneumothorax; nerve injury; bleeding complications; infection; or local anaesthetic systemic toxicity. This included at least 12 cases where blocks were performed in therapeutically anticoagulated patients [37, 40, 49, 54] and 105 cases where bilateral blocks or a combination of blocks were used [27, 39, 40, 43]. One report described keeping a serratus anterior plane block catheter in place for 7 days in a multitrauma patient with no signs of local or systemic infection [57].

Four RCTs specifically evaluated the haemodynamic effects of serratus anterior plane block. Two studies [13, 16] found no significant differences in haemodynamic parameters, including heart rate and mean arterial pressure, between serratus anterior plane block and systemic analgesia alone. In another study, Saad et al. [18] reported four episodes of hypotension in the group receiving a thoracic paravertebral block compared with none in the serratus anterior plane block group. Khalil et al. [20] compared continuous serratus anterior plane block with thoracic epidural analgesia and noted that mean arterial blood pressure was consistently lower in the thoracic epidural analgesia group. The thoracic epidural analgesia group also had significantly more episodes of hypotension requiring treatment. There is therefore level 2 evidence that the serratus anterior plane block is associated with a low risk of haemodynamic instability.

Discussion

The results of this systematic review indicate that investigations into serratus anterior plane and PECS blocks in cardiothoracic surgery and trauma are still at an early stage. We identified nine RCTs, of which seven studied serratus anterior plane block, one studied PECS 2 block and one compared both serratus anterior plane block and PECS 2 block with a control group. Eight of the nine RCTs evaluated analgesia after thoracotomy (n = 3) or video-assisted thoracoscopic surgery (n = 5), and one evaluated PECS 2 blocks post-sternotomy analgesia. The majority of the literature in other cardiothoracic applications comprises small case series and case reports (largely available only as conference abstracts) and thus serves primarily as proof-of-concept. These applications include: facilitation of tracheal

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extubation following thoracotomy and cardiopulmonary resuscitation-associated rib pain; transcatheter aortic valve replacement via the subclavian approach; cardiac device implantation; rib fractures and associated trauma; latissimus dorsi tendon transfer surgery; portacath insertion; and chest wall liposuction.

Nevertheless, the available evidence to date is encouraging for the efficacy of serratus anterior plane and PECS blocks in cardiothoracic surgery. They consistently improve early postoperative analgesia when added to systemic analgesia. The clinical duration of analgesia with single-injection techniques appears to be in the region of 6–12 h, with greatest analgesic benefit in the first 4–6 h. Analgesic duration is longer than that provided by multi-level intercostal nerve blocks [19, 34] but shorter when compared with single-injection thoracic paravertebral blockade [18]. However, when a continuous serratus anterior plane block technique is used, a single small RCT indicates that it provides analgesia similar to thoracic epidural analgesia following thoracotomy [20]. There are no studies comparing continuous serratus anterior plane block and continuous thoracic paravertebral blockade. Although continuous serratus anterior plane block may be easier and safer to perform compared with continuous thoracic epidural analgesia or paravertebral blockade, more high-quality studies are needed to determine if it provides equivalent analgesia in cardiothoracic surgery. The role of these blocks in minimally invasive cardiac procedures similarly warrants further exploration as their use may be associated with lower postoperative complication rates [44, 50].

With regard to median sternotomy, bilateral PECS 2 blocks were shown to be superior to systemic analgesia alone, although the mechanism of this is unclear. PECS blocks are not expected to block the anterior cutaneous branches of the intercostal nerves [3]; however, they may exert an analgesic effect via a reduction in pectoral or intercostal muscle spasm. It is uncertain how they compare with paraspinal fascial plane blocks, such as the erector spinae plane block, which are being increasingly used in cardiac surgery [64].

The evidence supporting the use of serratus anterior plane and PECS blocks in thoracic trauma is presently confined to small case series and case reports. However, it is reasonable to infer from the thoracic surgical data that they are likely to be effective alternatives to thoracic epidural analgesia and thoracic paravertebral blockade for anterolateral rib fractures, especially where these latter techniques are contraindicated. Deep serratus anterior plane blocks may even be effective in posterior rib fractures [51] as the associated soft tissue trauma may contribute to more extensive spread [65].

Efficacy aside, the serratus anterior plane and PECS blocks represent an attractive alternative to thoracic epidural analgesia and paravertebral blockade due to their relative simplicity and perceived safety. They can be performed in the supine position and under general anaesthesia if necessary, and carry little-to-no risk of pleural puncture or spinal cord injury. The margin of safety in anticoagulated patients is higher as the site of injection is relatively superficial, compressible and distant from structures vulnerable to an expanding haematoma. There is also a minimal risk of hypotension, which can be attributed to the more peripheral site of action and consequent lack of sympathectomy. The greater haemodynamic stability compared with thoracic epidural analgesia [20] and paravertebral blockade [18] is a particular advantage in cardiac patients with other co-morbidities.

At the same time, however, the lack of sympathetic blockade means that the serratus anterior plane and PECS blocks will provide only somatic and not visceral analgesia, which may impact analgesic efficacy. More importantly, inter-individual variability in effectiveness can also be expected given that the extent of spread, and consequently the mass of local anaesthetic reaching target nerves, is influenced by factors such as site and volume of injection. In a recent anatomical study of the PECS 1 block, injection at a point medial to the pectoral branch of the thoraco-acromial artery produced spread only to the medial and lateral pectoral nerves, whereas a more lateral site of injection produced additional spread to the intercostobrachial nerve in the axilla [3]. With regard to the serratus anterior plane block, both site and volume of injection influence which intercostal nerve territories will be blocked [66], and thus patients may experience pain from sites outside of the zone of coverage (e.g. thoracostomy tubes placed in the lower intercostal spaces). It remains unclear if there is a clinically significant difference between injections superficial or deep to the serratus anterior muscle [67, 68], and there are currently no published clinical trials comparing both variants.

There are several limitations that must be borne in mind when interpreting the findings of this review. First, these are relatively novel techniques and we were able to discover only a small number of RCTs involving just over 600 patients. The majority of the current literature is comprised of small cohort studies, case series and case reports. Although these are overwhelmingly positive, publication bias must be taken into account. In addition, many were published as

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conference proceedings rather than in peer-reviewed journals. Second, study heterogeneity, together with the limited evidence base, prevented us from performing a meaningful meta-analysis, and readers must thus make their own final inferences from the aggregation of presented data. Finally, many clinically relevant questions have yet to be adequately addressed. These include: the comparison of continuous fascial plane techniques with continuous thoracic paravertebral or epidural analgesia, which are still considered the gold standard for thoracic analgesia; the impact of optimal systemic multimodal analgesic regimens on the benefit offered by fascial plane blocks; and the effect of using different volumes and concentrations of local anaesthetic solutions and additives.

In conclusion, research into the use of serratus anterior plane and PECS blocks in cardiothoracic surgery and trauma is still at an early stage. Nevertheless, the current evidence indicates that these blocks may be a safe and effective option for thoracic analgesia, particularly where more invasive techniques such as thoracic epidural analgesia or thoracic paravertebral blockade are not feasible. Further, comparative trials are needed to determine which technique offers the optimal benefit-to-risk ratio in specific surgical settings.

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References

1. Chin KJ. Thoracic wall blocks: from paravertebral to retrolaminar to serratus to erector spinae and back again – A review of evidence. *Best Practice and Research Clinical Anaesthesiology* 2019; **33**: 67–77.
2. Chin KJ, Pawa A, Forero M, Adhikary S. Ultrasound-guided fascial plane blocks of the thorax: pectoral I and II, serratus anterior plane, and erector spinae plane blocks. *Advances in Anesthesia* 2019; **37**: 187–205.
3. Versyck B, Groen G, van Geffen GJ, Van Houwe P, Bleys RL. The pecs anesthetic blockade: a correlation between magnetic resonance imaging, ultrasound imaging, reconstructed cross-sectional anatomy and cross-sectional histology. *Clinical Anatomy* 2019; **32**: 421–9.
4. Versyck B, van Geffen GJ, Chin KJ. Analgesic efficacy of the Pecs II block: a systematic review and meta-analysis. *Anaesthesia* 2019; **74**: 663–73.
5. Higgins JPT, Thomas J, Chandler J, et al., eds. Cochrane handbook for systematic reviews of interventions version 6.0 (updated July 2019). Cochrane. 2019. www.training.cochrane.org/handbook (accessed 16/08/2019).
6. Higgins JP, Lasserson T, Chandler J, Tovey D, Churchill R. Methodological expectations of cochrane intervention reviews (MECIR). 2019. https://methods.cochrane.org/methodological-expectations-cochrane-intervention-reviews (accessed 01/11/2019).
7. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Medicine* 2009; **6**: e1000097.
8. Rethlefsen M, Koffel J, Kirtley S, Waffenschmidt S, Ayala AP. Preferred reporting items for systematic reviews and meta-analysis search extension (PRISMA-S) 2019: explanation and elaboration. 2019. https://osf.io/sfc38/ (accessed 14/05/2019).
9. McGowan J, Sampson M, Salzwedel DM, Cogo E, Foerster V, Lefebvre C. PRESS peer review of electronic search strategies: 2015 guideline statement. *Journal of Clinical Epidemiology* 2016; **75**: 40–6.
10. Higgins JPT, Altman DG, Gøtzsche PC, et al. The cochrane collaboration's tool for assessing risk of bias in randomised trials. *British Medical Journal* 2011; **343**: d5928.
11. Wells G, Shea B, O'Connell D et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. 2013. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp (accessed 17/11/2019).
12. Oxford Centre for Evidence-Based Medicine Levels of Evidence Working Group. The Oxford levels of evidence 2. 2011. https://www.cebm.net/index.aspx?o = 5653 (accessed 15/11/2019).
13. Kim DH, Oh YJ, Lee JG, Ha D, Chang YJ, Kwak HJ. Efficacy of ultrasound-guided serratus plane block on postoperative quality of recovery and analgesia after video-assisted thoracic surgery: a randomized, triple-blind, placebo-controlled study. *Anesthesia and Analgesia* 2018; **126**: 1353–61.
14. Lee J, Kim S. The effects of ultrasound-guided serratus plane block, in combination with general anesthesia, on intraoperative opioid consumption, emergence time, and hemodynamic stability during video-assisted thoracoscopic lobectomy: a randomized prospective study. *Medicine* 2019; **98**: e15385.
15. Ökmen K, Metin Okmen B. Evaluation of the effect of serratus anterior plane block for pain treatment after video-assisted thoracoscopic surgery. *Anaesthesia, Critical Care and Pain Medicine* 2018; **37**: 349–53.
16. Semyonov M, Fedorina E, Grinshpun J, et al. Ultrasound-guided serratus anterior plane block for analgesia after thoracic surgery. *Journal of Pain Research* 2019; **12**: 953–60.
17. Park MH, Kim JA, Ahn HJ, Yang MK, Son HJ, Seong BG. A randomised trial of serratus anterior plane block for analgesia after thoracoscopic surgery. *Anaesthesia* 2018; **73**: 1260–4.
18. Saad FS, El Baradie SY, Aliem M, Ali MM, Kotb TAM. Ultrasound-guided serratus anterior plane block versus thoracic paravertebral block for perioperative analgesia in thoracotomy. *Saudi Journal of Anaesthesia* 2018; **12**: 565–70.
19. Kaushal B, Chauhan S, Saini K, et al. Comparison of the efficacy of ultrasound-guided serratus anterior plane block, pectoral nerves ii block, and intercostal nerve block for the management of postoperative thoracotomy pain after pediatric cardiac surgery. *Journal of Cardiothoracic and Vascular Anesthesia* 2019; **33**: 418–25.
20. Khalil AE, Abdallah NM, Bashandy GM, Kaddah TAH. Ultrasound-guided serratus anterior plane block versus thoracic epidural analgesia for thoracotomy pain. *Journal of Cardiothoracic and Vascular Anesthesia* 2017; **31**: 152–8.
21. Kumar KN, Kalyane RN, Singh NG, et al. Efficacy of bilateral pectoralis nerve block for ultrafast tracking and postoperative pain management in cardiac surgery. *Annals of Cardiac Anaesthesia* 2018; **21**: 333–8.
22. Berthoud V, Ellouze O, Nguyen M, et al. Serratus anterior plane block for minimal invasive heart surgery. *BMC Anesthesiology* 2018; **18**: 144.
23. Okmen K, Okmen BM. The efficacy of serratus anterior plane block in analgesia for thoracotomy: a retrospective study. *Journal of Anesthesia* 2017; **31**: 579–85.

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24. Munshey F, Ramamurthi RJ, Tsui B. Early experience with PECS 1 block for Port-a-Cath insertion or removal in children at a single institution. *Journal of Clinical Anesthesia* 2018; **49**: 63–4.
25. Arumugam S, Kanelos P, Knauft W, Thayer J, Sinha S. Comparison of serratus plane block with thoracic epidural for post-operative analgesia in thoracic wall surgeries-a retrospective analysis. In: Spring 2016 Abstract Titles: aSRA 41st Annual Regional Anesthesiology and Acute Pain Medicine Meeting March 31 – April 2, 2016, New Orleans, LA. *Regional Anesthesia and Pain Medicine* 2016; **41**: 650.
26. de Waroux JBL, Gauthey A, Chehab M, et al. Echoguided serratus anterior plane block for intermuscular implantation of the subcutaneous implantable cardioverter-defibrillator: a feasibility study. *Acta Cardiologica* 2017; **72**: 574–574.
27. Marcoe GP, Wood H, Marcoe JP, Irfan FB. Multimodal analgesia and regional anesthesia in cardiac surgery. *Clinical Nutrition ESPEN* 2019; **31**: 121.
28. Monico C. Pectoralis (PECS) I/II block for intracardiac device placement in a Large Pediatric Heart Center. In: Scientific Abstracts and ePosters: ASRA 42nd Annual Regional Anesthesiology and Acute Pain Medicine Meeting April 6–8, 2017 San Francisco, CA. *Regional Anesthesia and Pain Medicine* 2017; **42**: 806.
29. Hillermann C, Millerchip S. Rib fracture management: comparing 3 different regional anaesthetic techniques. In: Abstracts and Highlight Papers of the 36th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2017. *Regional Anesthesia and Pain Medicine* 2017; **42** (Suppl. 1): e84.
30. Moll V, Maffeo C, Mitchell M, et al. Association of serratus anterior plane block for minimally invasive direct coronary artery bypass surgery with higher opioid consumption: a retrospective observational study. *Journal of Cardiothoracic and Vascular Anesthesia* 2018; **32**: 2570–7.
31. Öksüz G, Sayan M. Analgesic effectiveness of serratus anterior block for video-assisted thoracoscopic surgery. *Journal of Cardio-Vascular-Thoracic Anaesthesia and Intensive Care Society* 2019; **25**: 17–22.
32. Tore Altun G, Arslantas MK, Corman Dincer P, Aykac ZZ. Ultrasound-guided serratus anterior plane block for pain management following minimally invasive repair of pectus excavatum. *Journal of Cardiothoracic and Vascular Anesthesia* 2019; **33**: 2487–91.
33. Cole N, T K, Pelletier M, Shook D, Nyman C, Vlassakov K. Regional anesthesia for the awake TAVR: a case series. In: abstracts and highlight papers of the 36th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2017. *Regional Anesthesia and Pain Medicine* 2017; **42**: e145.
34. Öksüz G, Sayan M, Arslan M, et al. The comparison of serratus anterior plane block versus intercostal block for postoperative analgesia following thoracotomy surgery. *Anestezi Dergisi* 2018; **26**: 223–8.
35. Kumar V, Gaber M, Moore F, Castresana MR. Pectoral blocks as an alternative for pain management after video assisted thoracoscopic surgeries and its potential role in enhancing postoperative recovery. *Anesthesia and Analgesia* 2016; **122** (Suppl. 3): S325.
36. Arumugam S, Kanelos P, Knauft W, Thayer J., Sinha S. Serratus plane block for post-operative analgesia in thoracic wall surgeries. In: Spring 2015 Abstract Titles: ASRA 40th Annual Regional Anesthesiology and Acute Pain Medicine Meeting May 14–16, 2015, Las Vegas, NV. *Regional Anesthesia and Pain Medicine* 2015; **40**: 644.
37. Barbera C, Milito P, Punturieri M, Asti E, Bonavina L. Serratus anterior plane block for hybrid transthoracic esophagectomy: a pilot study. *Journal of Pain Research* 2017; **10**: 73–7.
38. Bunnell A, Bebic Z, Hagen J. Serratus anterior block as a novel approach to postoperative pain control after pediatric thoracotomy: a case report. In: Scientific Abstracts and ePosters: ASRA 42nd Annual Regional Anesthesiology and Acute Pain Medicine Meeting April 6–8, 2017, San Francisco, CA. *Regional Anesthesia and Pain Medicine* 2017; **42**: 802–18.
39. Schuitemaker Requena JB, Arteaga Mejia D, Lopez Pantaleon LA, Mayoral Ripamonti JT, Imbiscuso Esqueda AT, Sanchez Cohen AP. Clavicular and ribs fractures a new approach for analgesia: PECII plus serratus plane block, report of 2 cases. In: Abstracts and Highlight Papers of the 35th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2016. *Regional Anesthesia and Pain Medicine* 2016; **41** (Suppl. 1): e93.
40. Watts AJ, Tong D. Serratus anterior plane blocks for pain control in anticoagulated patients with traumatic rib fractures. A case series. In: Abstracts and Highlight Papers of the 37th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2018. *Regional Anesthesia and Pain Medicine* 2018; **43** (Suppl. 1): e142.
41. Bin WH, Lim V, Tan LZ, Tan JCH. Case series: serratus anterior plane block (SAPB) for latissimus dorsi tendon transfer surgery. In: Abstracts and Highlight Papers of the 35th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2016. *Regional Anesthesia and Pain Medicine* 2016; **41** (Suppl. 1): e63.
42. Garcia-Verdugo Peralta O, Tielens L, Bruhn J. Modified serratus plane block for pain management in chest wall surgery in children: a preliminary study. *Anesthesia and Analgesia* 2016; **123**(Suppl. 2): 312.
43. Prenter A, Quinn J, Raj N. A prosPECtive technique. *Anaesthesia* 2015; **70**: 11.
44. Block M, Pitchon DN, Schwenk ES, Ruggiero N, Entwistle J, Goldhammer JE. Left subclavian transcatheter aortic valve replacement under combined interscalene and pectoralis nerve blocks: a case series. *AandA Practice* 2018; **18**: 18.
45. Droghetti A, Basso Ricci E, Scimia P, Harizai F, Marini M. Ultrasound-guided serratus anterior plane block combined with the two-incision technique for subcutaneous ICD implantation. *Pacing and Clinical Electrophysiology* 2018; **41**: 517–23.
46. Johnson M, Doyle Y, McNeill D, McGinley J. A case series of pectoralis blocks in the paediatric cath lab; a novel technique for postoperative analgesia. *Regional Anesthesia and Pain Medicine* 2015; **40**: e196.
47. Foley E, Bashir F, Gaber M, Kumar V, Castresana M. Ultrasound-guided serratus plane block analgesia for blunt thoracic trauma: a series of nine cases. *Critical Care Medicine* 2018; **46**: 780.
48. Parras T, Malik Z. Ng L, Duran J. Case series: serratus plane block for rib fracture analgesia. In: Abstracts and Highlight Papers of the 35th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2016. *Regional Anesthesia and Pain Medicine* 2015; **40** (Suppl. 1): e118.
49. Kumar V, Gaber M, Moore F, Castresana M. Ultrasound guided pectoral nerve block (PEC 1 and 2) as an alternative for pain management in patients with multiple rib fractures. In: Spring 2016 Abstract Titles: ASRA 41st Annual Regional Anesthesiology and Acute Pain Medicine Meeting March 31 – April 2, 2016, New Orleans, LA. *Regional Anesthesia and Pain Medicine* 2016; **41**: 651.
50. Miller MA, Bhatt HV, Weiner M, et al. Implantation of the subcutaneous implantable cardioverter-defibrillator with truncal plane blocks. *Heart Rhythm* 2018; **15**: 1108–11.
51. Rose P, Ramlogan R, Sullivan T, Lui A. Serratus anterior plane blocks provide opioid-sparing analgesia in patients with

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isolated posterior rib fractures: a case series. *Canadian Journal of Anesthesia* 2019; **66**: 1263–4.
52. Joyce A, Ivie R, Ketchandji D. Use of subpectoral interfascial plane blocks to reduce ventilatory requirements following cardiopulmonary resuscitation: a case series. *Pain Medicine (United States)* 2019; **20**: 609–10.
53. Jenkins H, Abdlaziz I, Kadry M. Audit on guidelines for management of patients with multiple rib fractures. *Anaesthesia* 2018; **73**(Suppl. 4): 79.
54. George RM, Yared M, Wilson SH. Deep serratus plane catheter for management of acute postthoracotomy pain after descending aortic aneurysm repair in a morbidly obese patient: a case report. *Anesthesia and Analgesia Practice* 2018; **10**: 49–52.
55. Teniola B, Mitchell M, Bansal V. Serratus anterior plane catheter for acute thoracotomy pain in a pediatric patient. In: Scientific Abstracts and ePosters: ASRA 42nd Annual Regional Anesthesiology and Acute Pain Medicine Meeting April 6–8, 2017, San Francisco, CA. *Regional Anesthesia and Pain Medicine* 2017; **42**: 802–18.
56. Sahin A, Baran O. Relocation of an infected cardiac pacemaker generator under ultrasound-guided pectoralis nerve block: a case study. *AandA practice* 2019; **12**: 285–7.
57. Abouelmagd R, Kumar A, Stack C. Bilateral serratus plane blocks; A novel approach to bilateral rib fracture pain control in a polytrauma patient. In: Abstracts and Highlight Papers of the 36th Annual European Society of Regional Anaesthesia and Pain Therapy (ESRA) Congress 2017. *Regional Anesthesia and Pain Medicine* 2017; **42**: e65–6.
58. Arunachalam S, Townsley P, Bedforth N. Pecs and serratus plane block for chest wall analgesia in multiple rib fractures. *Anaesthesia* 2014; **69**: 11.
59. Mbuvah F, Garzon-Serrano J. Ultrasound guided serratus anterior plane block for thoracic analgesia in a patient with multiple rib fractures. In: Scientific Abstracts and ePosters: ASRA 42nd Annual Regional Anesthesiology and Acute Pain Medicine Meeting April 6–8, 2017, San Francisco, CA. *Regional Anesthesia and Pain Medicine* 2017; **42**: 802–18.
60. Moll V, Groff R, Budhrani G, Sathiyakumar A, McKenzie-Brown AM. Bilateral serratus anterior plane block facilitates ventilator weaning in patient with rib cage pain. *Critical Care Medicine* 2016; **44**: 561.
61. Moore F, Kumar V, Gaber M, Castresana M. Continuous PEC 2 analgesia in a trauma patient with multiple rib fractures and contraindication to or failure of other modes of analgesia. In: Spring 2016 Abstract Titles: ASRA 41st Annual Regional Anesthesiology and Acute Pain Medicine Meeting March 31 – April 2, 2016, New Orleans, LA. *Regional Anesthesia and Pain Medicine* 2016; **41**: 640.
62. Wong M, DeHaan J, Hernandez N, Erian M. The serratus plane block: challenging the gold standard for multiple rib fractures. In: Scientific Abstracts and ePosters: ASRA 42nd Annual Regional Anesthesiology and Acute Pain Medicine Meeting April 6–8, 2017, San Francisco, CA. *Regional Anesthesia and Pain Medicine* 2017; **42**: 802–18.
63. Rose P, Ramlogan R, Madden S, Lui A. Serratus anterior plane block home catheter for posterior rib fractures and flail chest. *Canadian Journal of Anesthesia* 2019; **66**: 997–8.
64. Noss C, Anderson KJ, Gregory AJ. Erector spinae plane block for open-heart surgery: a potential tool for improved analgesia. *Journal of Cardiothoracic and Vascular Anesthesia* 2019; **33**: 376–7.
65. Johnston DF, Black ND, O'Halloran R, Turbitt LR, Taylor SJ. Cadaveric findings of the effect of rib fractures on spread of serratus plane injections. *Canadian Journal of Anesthesia* 2019; **66**: 738–9.
66. Biswas A, Castanov V, Li Z, et al. Serratus plane block: a cadaveric study to evaluate optimal injectate spread. *Regional Anesthesia and Pain Medicine* 2018; **43**: 854–8.
67. Abdallah FW, Cil T, MacLean D, et al. Too deep or not too deep? A propensity-matched comparison of the analgesic effects of a superficial versus deep serratus fascial plane block for ambulatory breast cancer surgery. *Regional Anesthesia and Pain Medicine* 2018; **43**: 480–7.
68. Piracha MM, Thorp SL, Puttanniah V, Gulati A. "A tale of two planes": deep versus superficial serratus plane block for postmastectomy pain syndrome. *Regional Anesthesia and Pain Medicine* 2017; **42**: 259–62.

Supporting Information

Additional supporting information may be found online via the journal website.

Table S1. Randomised controlled trials and retrospective cohort studies comparing pectoral nerves (PECS) blocks and serratus anterior plane blocks (SAPB) to comparators: primary outcomes.

Table S2. Methodological quality of non-randomised studies using the Newcastle-Ottawa Scale.

Table S3. Non-randomised studies of pectoral nerves (PECS) 1 and 2 and serratus anterior plane blocks in thoracotomy or video-assisted thoracoscopic surgery.

Table S4. Non-randomised studies on pectoral nerves (PECS) 1 and 2 and serratus anterior plane blocks in cardiac-related interventional procedures.

Table S5. Non-randomised studies on pectoral nerves (PECS) 1 and 2 and serratus anterior plane blocks in thoracic trauma.

Table S6. Non-randomised studies of the pectoral nerves (PECS) 1 and 2 and serratus anterior plane blocks in miscellaneous applications of the anterolateral chest wall.

Appendix S1: Search strategies.

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